Handle Control

1. Course Content

1. Control the robot's movement using a PS2 handle. After running the program, use the PS2 handle to control the robot chassis.

2. Preparations

2.1 Content Description

This lesson uses the Jetson Orin NX as an example. For Raspberry Pi boards, you need to open a terminal on the host machine, enter the command to get into the Docker container, and then input the commands mentioned in this lesson in the terminal within the Docker container. A tutorial on how to enter the Docker container from the host can be found in this product's guide [00.Configuration and Operation Guide]-[3.Enter the Docker (For RPI5)]. For ORIN and RDK boards, you can directly open a terminal and enter the commands mentioned in this lesson.

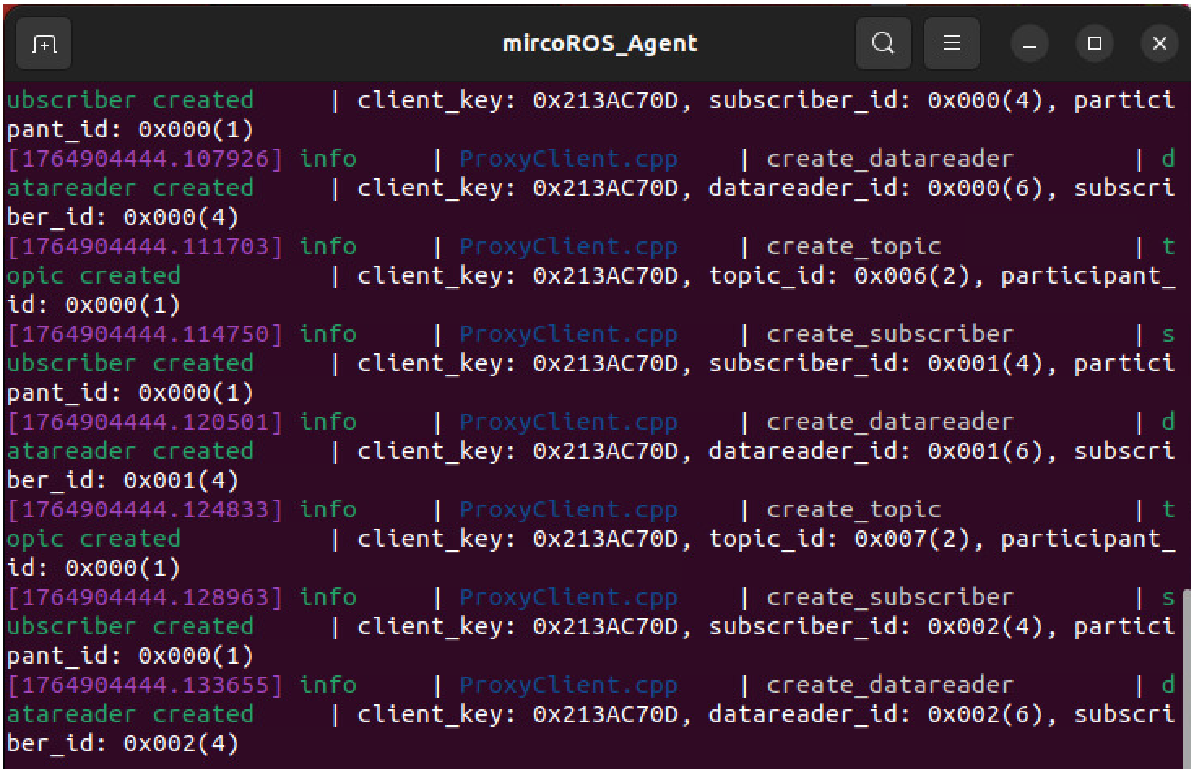
2.2 Start the Agent

Note: To test all examples, you must start the Docker agent first. If it is already running, you do not need to start it again.

Enter the command in the vehicle's terminal:

```
start_agent
```

The terminal will print the following information, indicating a successful connection:



```
ubscriber created     | client_key: 0x213AC70D, subscriber_id: 0x000(4), partici
pant_id: 0x000(1)
[1764904444.107926] info     | ProxyClient.cpp    | create_datareader        | d
atareader created     | client_key: 0x213AC70D, datareader_id: 0x000(6), subscri
ber_id: 0x000(4)
[1764904444.111703] info     | ProxyClient.cpp    | create_topic             | t
opic created          | client_key: 0x213AC70D, topic_id: 0x006(2), participant_
id: 0x000(1)
[1764904444.114750] info     | ProxyClient.cpp    | create_subscriber        | s
ubscriber created     | client_key: 0x213AC70D, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1764904444.120501] info     | ProxyClient.cpp    | create_datareader        | d
atareader created     | client_key: 0x213AC70D, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1764904444.124833] info     | ProxyClient.cpp    | create_topic             | t
opic created          | client_key: 0x213AC70D, topic_id: 0x007(2), participant_
id: 0x000(1)
[1764904444.128963] info     | ProxyClient.cpp    | create_subscriber        | s
ubscriber created     | client_key: 0x213AC70D, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1764904444.133655] info     | ProxyClient.cpp    | create_datareader        | d
atareader created     | client_key: 0x213AC70D, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
```

2.3 Check the Device

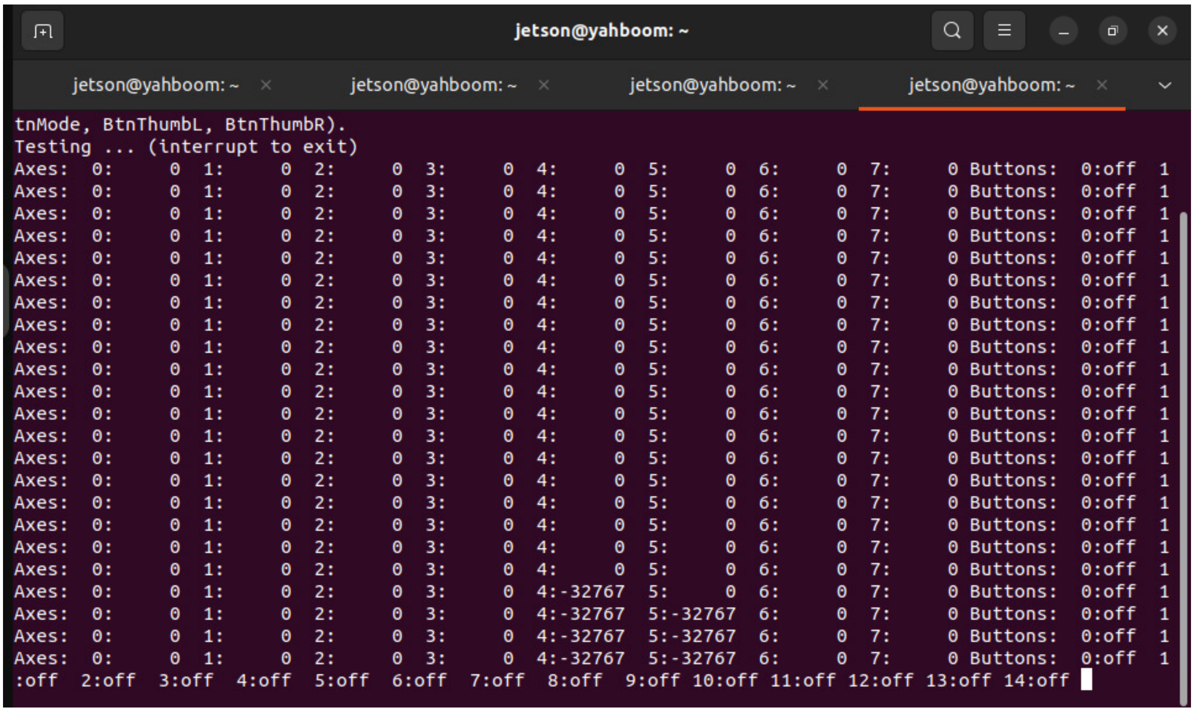
First, insert the wireless handle's USB receiver into the main controller, open a terminal, and enter the following command. If `js0` is displayed, this is the wireless handle. In special cases, you can also check the device list changes by connecting and disconnecting the wireless handle's USB receiver. If there is a change, the changing device is the correct one; otherwise, the connection is unsuccessful or the device is not recognized.

2.4 Test Handle Input

> [!TIP]
>
> - Testing the handle input here is for checking key value changes and is used for debugging. You can skip this step for normal use.

Open a terminal and enter the following command. As shown in the figure, the wireless handle has 8 analog stick inputs and 15 button inputs. You can press the buttons to test the corresponding numbers.

```
sudo jstest /dev/input/js0
```



```
tnMode, BtnThumbL, BtnThumbR).
Testing ... (interrupt to exit)
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:     0  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:-32767  5:     0  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:-32767  5:-32767  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:-32767  5:-32767  6:     0  7:     0 Buttons:  0:off  1
Axes:  0:     0  1:     0  2:     0  3:     0  4:-32767  5:-32767  6:     0  7:     0 Buttons:  0:off  1
:off  2:off  3:off  4:off  5:off  6:off  7:off  8:off  9:off 10:off 11:off 12:off 13:off 14:off
```

3. Run the Example

3.1 Start the PS2 Handle Control Node

For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first.

```
bringup_m3
```

Open a terminal inside the container,

```
exec_m3
```

Shortcut command:

```
ps2
```

Full launch command:

```
ros2 launch m3_bringup joy_contoller_launch.py
```

3.2 Button Control Description

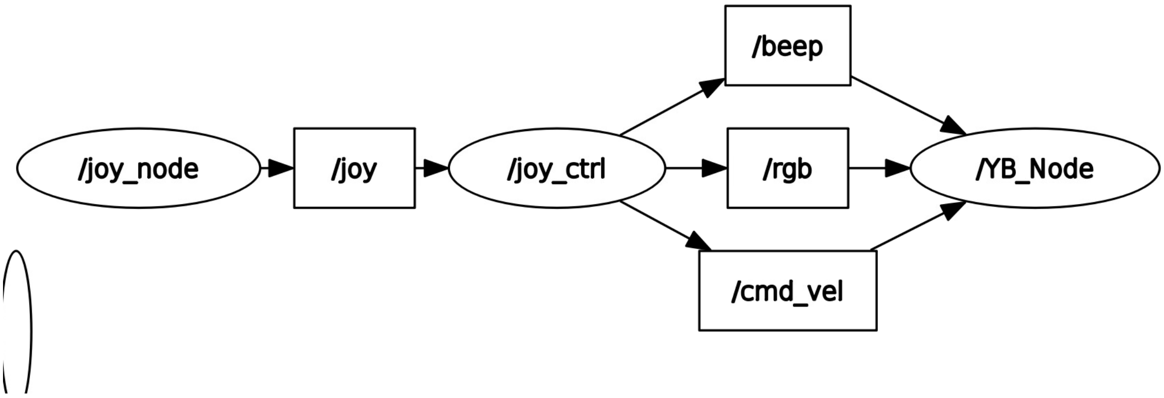
Handle Action	Function
Left Stick Up/Down	Car moves forward/backward
Left Stick Left/Right	Car strafes left/right
Right Stick Left/Right	Car rotates left/right
"START" Button	Exit sleep/Control buzzer
Left Stick Press	Adjust X/Y axis linear speed
Right Stick Press	Adjust angular speed
Right Button 1	Switch ambient light mode

4. Source Code Analysis

4.1 View Communication Between Nodes

- Open a terminal and enter the command:

```
ros2 run rqt_graph rqt_graph
```



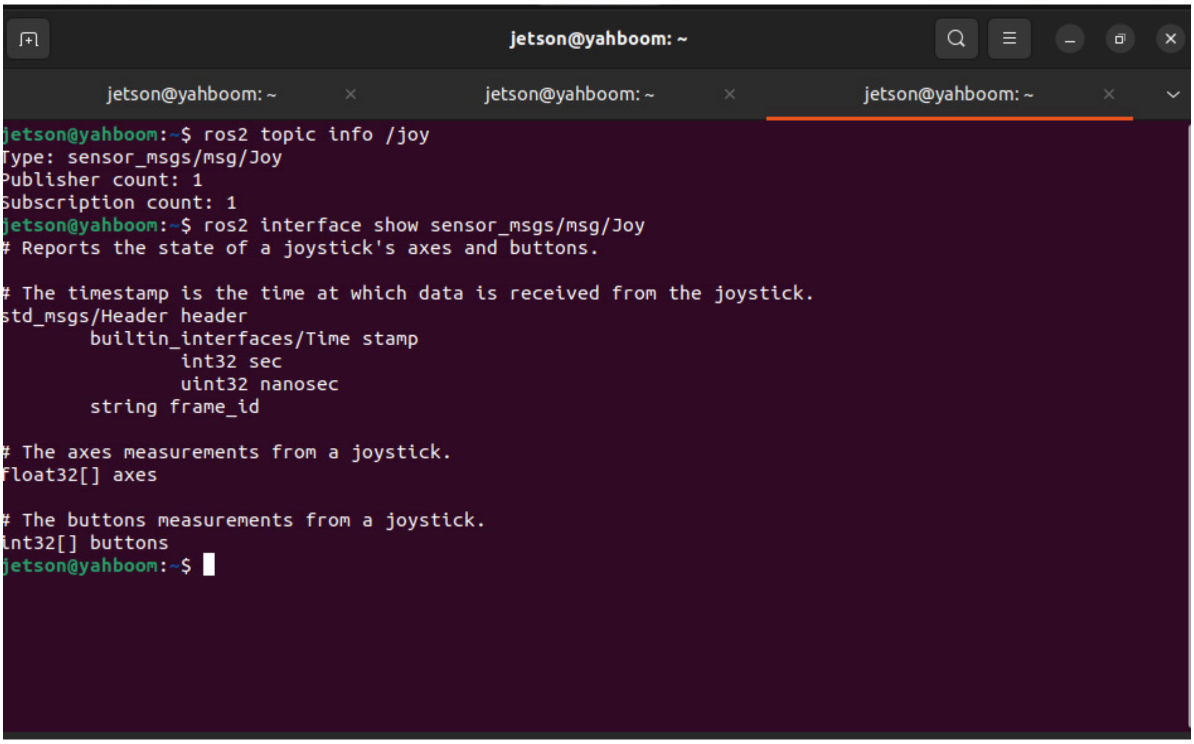
In the node relationship diagram above:

- `joy_node`: Gets data from the handle receiver and publishes it to the `/joy` topic.
- `joy_ctrl`: Subscribes to the data on the `/joy` topic, parses the buttons and corresponding actions, and publishes to the `/cmd_vel` and `/rgb` topics to control the robot chassis.

4.2 View Topics and Message Types

Open a terminal on the vehicle or virtual machine and enter the command:

```
ros2 interface show sensor_msgs/msg/Joy
```



```
jetson@yahboom:~$ ros2 topic info /joy
Type: sensor_msgs/msg/Joy
Publisher count: 1
Subscription count: 1
jetson@yahboom:~$ ros2 interface show sensor_msgs/msg/Joy
# Reports the state of a joystick's axes and buttons.

# The timestamp is the time at which data is received from the joystick.
std_msgs/Header header
        builtin_interfaces/Time stamp
                int32 sec
                uint32 nanosec
        string frame_id

# The axes measurements from a joystick.
float32[] axes

# The buttons measurements from a joystick.
int32[] buttons
jetson@yahboom:~$ 
```

As you can see, the data on the `/joy` topic is a timestamped array of float32 numbers.

- `float32[] axes`: 8 analog stick input data.
- `int32[] buttons`: 15 button inputs.

4.3 Source Code Analysis

Source code path:

```
~/yahboomM3_ws/code_ws/src/m3_common_demo/m3_common_demo/ps2_controller/joy2ros.
py
```

4.3.1 Topic Communication

```
self.pub_cmdVel = self.create_publisher(Twist,'cmd_vel',  1)
self.sub_Joy = self.create_subscription(Joy,'joy', self.buttonCallback,10)
```

Here, the chassis movement is controlled by publishing to the `/cmd_vel` topic, and the PS2 handle's button states are obtained by subscribing to the `/joy` topic.

The basic principle of controlling the robot with a handle is to first parse the handle data, then convert it into corresponding control state quantities, and send them to the chassis control board via topic communication. The chassis control board subscribes to the topic data and converts it into values for controlling the hardware.

4.3.2 Chassis Movement Control

The PS2 handle controls the chassis movement through the `user_jetson` method in the `JoyTeleop` class:

```python
class JoyTeleop(Node):
    def __init__(self, name):
        super().__init__(name)
        self.Joy_active = True
        self.Buzzer_active = 0
        self.RGBLight_index = 0
        self.cancel_time = time.time()
        self.user_name = getpass.getuser()
        self.linear_Gear = 1.0 / 3
        self.angular_Gear = 1.0 / 4

        self.loop_active = True

        # create pub
        self.pub_cmdVel = self.create_publisher(Twist, "cmd_vel", 1)
        self.pub_Buzzer = self.create_publisher(UInt16, "/beep", 1)
        self.pub_JoyState = self.create_publisher(Bool, "JoyState", 10)
        self.pub_RGBLight = self.create_publisher(ColorRGBA, "rgb", 10)

        # create sub
        self.sub_Joy = self.create_subscription(Joy, "joy", self.buttonCallback,
10)

        # declare parameter and get the value
        self.declare_parameter("xspeed_limit", 1.0)
        self.declare_parameter("yspeed_limit", 1.0)
        self.declare_parameter("angular_speed_limit", 5.0)
        self.xspeed_limit = (

 self.get_parameter("xspeed_limit").get_parameter_value().double_value
        )
        self.yspeed_limit = (

 self.get_parameter("yspeed_limit").get_parameter_value().double_value
        )
        self.angular_speed_limit = (

 self.get_parameter("angular_speed_limit").get_parameter_value().double_value
        )

    def buttonCallback(self, joy_data):
        if not isinstance(joy_data, Joy):
            return
        self.user_jetson(joy_data)


    def user_jetson(self, joy_data):
        # arm_ctrl_start
        if joy_data.buttons[10] == 1:
            pass
        if (
            joy_data.buttons[0]
```

```
            == joy_data.buttons[1]
            == joy_data.buttons[6]
            == joy_data.buttons[3]
            == joy_data.buttons[4]
            == 0
            and joy_data.axes[7] == joy_data.axes[6] == 0
            and joy_data.axes[5] != -1
        ):
            self.loop_active = False
        else:
            if joy_data.buttons[3] == 1:
                print("1,-")
                # X
            if joy_data.buttons[1] == 1:
                 # B
                print("1,+")
            if joy_data.buttons[0] == 1:
                # A
                print("2,-")
            if joy_data.buttons[4] == 1:
                # Y
                print("2,+")
            if joy_data.axes[6] != 0:
                 # Left button: left positive, right negative
                print("3,-/+")
            if joy_data.axes[7] != 0:
                 # Left button: up positive, down negative
                print("4,-/+")

        if joy_data.buttons[9] == 1:
            pass
        # RGBLight
        if joy_data.buttons[7] == 1:
            self.RGBLight_index = self.RGBLight_index + 1.0
            if self.RGBLight_index > 7:
                self.RGBLight_index = 0.0

            rgb__msg = ColorRGBA()
            rgb__msg.a = 100.0 + self.RGBLight_index
            self.pub_RGBLight.publish(rgb__msg)
        # Buzzer
        if joy_data.buttons[11] == 1:
            Buzzer_ctrl = UInt16()
            if self.Buzzer_active == 0:
                self.Buzzer_active = self.Buzzer_active + 1
            else:
                self.Buzzer_active = self.Buzzer_active - 1
            Buzzer_ctrl.data = self.Buzzer_active
            self.pub_Buzzer.publish(Buzzer_ctrl)
            # linear Gear control
        if joy_data.buttons[13] == 1:
            if self.linear_Gear == 1.0:
                self.linear_Gear = 1.0 / 3
            elif self.linear_Gear == 1.0 / 3:
                self.linear_Gear = 2.0 / 3
            elif self.linear_Gear == 2.0 / 3:
                self.linear_Gear = 1
            # angular Gear control
```

```
        if joy_data.buttons[14] == 1:
            if self.angular_Gear == 1.0:
                self.angular_Gear = 1.0 / 4
            elif self.angular_Gear == 1.0 / 4:
                self.angular_Gear = 1.0 / 2
            elif self.angular_Gear == 1.0 / 2:
                self.angular_Gear = 3.0 / 4
            elif self.angular_Gear == 3.0 / 4:
                self.angular_Gear = 1.0
        xlinear_speed = (
            self.filter_data(joy_data.axes[1]) * self.xspeed_limit *
self.linear_Gear
        )

        ylinear_speed = (
            self.filter_data(joy_data.axes[0]) * self.yspeed_limit *
self.linear_Gear
        )
        angular_speed = (
            self.filter_data(joy_data.axes[2])
            * self.angular_speed_limit
            * self.angular_Gear
        )
        if xlinear_speed > self.xspeed_limit:
            xlinear_speed = self.xspeed_limit
        elif xlinear_speed < -self.xspeed_limit:
            xlinear_speed = -self.xspeed_limit
        if ylinear_speed > self.yspeed_limit:
            ylinear_speed = self.yspeed_limit
        elif ylinear_speed < -self.yspeed_limit:
            ylinear_speed = -self.yspeed_limit
        if angular_speed > self.angular_speed_limit:
            angular_speed = self.angular_speed_limit
        elif angular_speed < -self.angular_speed_limit:
            angular_speed = -self.angular_speed_limit
        twist = Twist()
        twist.linear.x = xlinear_speed
        twist.linear.y = ylinear_speed
        twist.angular.z = angular_speed
        if self.Joy_active == True:
            self.pub_cmdVel.publish(twist)
    def filter_data(self, value):
        if abs(value) < 0.2:
            value = 0
        return value
```